Introduction to Ordinary Differential Equations

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Contents

1. Introduction **3**
1.1. Equations Containing No Derivatives, and Differential Equations 3
1.2. Static Viewpoint and Dynamic Viewpoint 4
1.3. Examples 8
1.4. Classification of Differential Equations 16

2. First-Order Differential Equations **21**
2.1. Linear Differential Equations; Method of Integrating Factors 21
2.2. Separable Differential Equations 26
2.3. Modeling with First-Order Differential Equations 30

1. Introduction

Lecture 1

§1.1. Equations Containing No Derivatives, and Differential Equations

9/07–9/10

Lecture material; cf. Boyce §1.1

Before saying what a differential equation *is*, it pays to say what it is *not*.

Definition 1.1.1 (Equation containing no derivatives)**.** An **algebraic equation** (more generally, an equation containing no derivatives) is a relation among numbers, such as

$$x^2 - 5x + 6 = 0 \qquad \text{or} \qquad \begin{cases} 2x + 3y = 7, \\ x - y = 1. \end{cases}$$

To *solve* it is to find the numbers — here $x \in \{2, 3\}$, and $(x, y) = (2, 1)$ — that make it true.

Definition 1.1.2 (Differential equation)**.** A **differential equation** is an equation relating an unknown *function* to one or more of its derivatives, such as

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y.$$

To *solve* it is to find the *functions* — here $y(t) = ce^t$ for any constant c — that make it true on some interval.

An algebraic equation asks *which number?* A differential equation asks *which function?* The unknown has been promoted from a point to a whole curve.

Observation . This promotion changes the shape of the answer. The solution set of a polynomial equation is a finite list of numbers; the solution set of a differential equation is typically an infinite *family* of functions, indexed by one or more arbitrary constants. The reason is that solving a differential equation involves integrating, and every integration brings with it a constant of integration.

Example 1.1.3. Compare:

	$x^2 - 5x + 6 = 0$	$y' = y$
unknown	a number x	a function $y(\cdot)$
solution set	$\{2, 3\}$	$\{\, ce^t : c \in \mathbb{R} \,\}$
how to pin it down	— (already finite)	one extra condition, e.g. $y(0) = 1$

Definition 1.1.4 (Initial condition, initial value problem)**.** A condition of the form $y(t_0) = y_0$ is an **initial condition**. A differential equation together with an initial condition is an **initial value problem** (IVP). Geometrically, the initial condition selects the one curve of the family that passes through the point (t_0, y_0).

Note 1.1.5 (Historical background I: Newton, Leibniz, the Bernoullis)**.** The subject originated with the calculus of Isaac Newton (1643–1727) and Gottfried Wilhelm Leibniz (1646–1716). Newton identified three forms of first-order equation, $\mathrm{d}y/\mathrm{d}x = f(x)$, $\mathrm{d}y/\mathrm{d}x = f(y)$ and $\mathrm{d}y/\mathrm{d}x = f(x, y)$, solving the last by infinite series when f is a polynomial. Leibniz gave us the notation $\mathrm{d}y/\mathrm{d}x$ and the integral sign, and discovered separation of variables (1691) and the solution of first-order linear equations (1694). The brothers Jakob (1654–1705) and Johann Bernoulli (1667–1748) of Basel developed the methods further; the brachistochrone problem, solved by both (and by Leibniz, Newton and l'Hôpital), was a famous source of friction between them.

§1.2. Static Viewpoint and Dynamic Viewpoint

Lecture material; not in Boyce

The whole subject rests on one change of perspective.

Static viewpoint: describe a quantity by its *value*.
Dynamic viewpoint: describe a quantity by its *rate of change*, together with where it started.

The second description is usually the one we can actually obtain, and — by the fundamental theorem of calculus — it contains exactly as much information as the first. Five illustrations.

§1.2.i. Five Illustrations

1. Salary

Two offers: company A starts you at 60k with a raise of 2k per year; company B starts you at 40k with a raise of 5k per year. Statically A is better. Dynamically B is better, and after

$$60 + 2t = 40 + 5t \Leftrightarrow t = \frac{20}{3} \approx 6.7 \text{ years}$$

B overtakes A and never looks back.

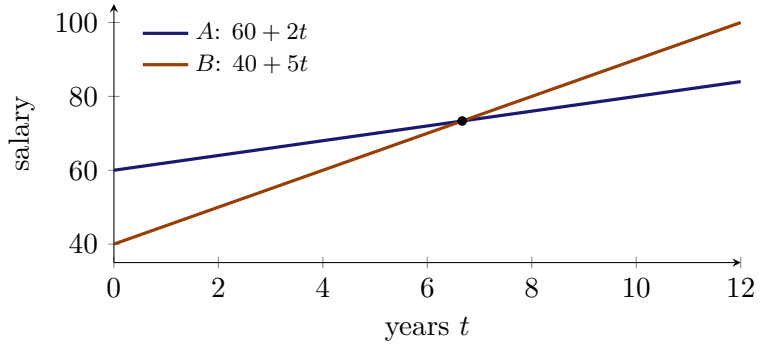


Figure 1.1.: The static comparison (values at $t = 0$) and the dynamic comparison (slopes) give opposite answers.

Over any horizon long enough to matter, the *derivative* beats the *value*.

2. Friends and careers

A popular slogan says that you become the average of the people you spend your time with. Take it literally: let $x(t)$ measure some trait of yours and let m be the average of that trait over the people around you. The slogan is not the statement $x = m$ (that is false at every instant); it is the statement that you *drift towards* m at a rate proportional to how far away you are:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = k\,(m - x), \qquad k > 0. \tag{1.1}$$

Observation . Equation (1.1) is literally Newton's law of cooling (subsection 1.3.ii), with your environment playing the role of the ambient temperature. Its solution is $x(t) = m + (x_0 - m)e^{-kt}$: the starting value x_0 is forgotten exponentially fast, and the only thing that survives in the long run is m. So a dynamic model says something a static one cannot — *who you are around eventually matters more than where you started*, and the rate k decides how fast "eventually" arrives.

3. The fundamental theorem of calculus

The dictionary translating between the two viewpoints is the fundamental theorem of calculus:

$$F(b) - F(a) = \int_a^b F'(t)\mathrm{d}t.$$

Intuition . Read from right to left, this says: *knowing the rate everywhere, plus one value, recovers the function everywhere.* In other words, the fundamental theorem of calculus is exactly the solution of the simplest possible initial value problem,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t), \quad y(t_0) = y_0 \qquad \Longrightarrow \qquad y(t) = y_0 + \int_{t_0}^t f(s)\mathrm{d}s.$$

Everything in this course is an attempt to do the same thing when the right-hand side is allowed to depend on y as well as on t.

4. Inequality proofs via derivatives

To prove $f(x) \geq g(x)$ for all $x \geq a$, it suffices to check the inequality *once*, at $x = a$, and then compare *rates*.

Proposition 1.2.1

Let f, g be differentiable on $[a, \infty)$. If $f(a) \geq g(a)$ and $f'(x) \geq g'(x)$ for all $x > a$, then $f(x) \geq g(x)$ for all $x \geq a$.

Proof. Let $h = f - g$. Then $h(a) \geq 0$ and $h' \geq 0$ on (a, ∞), so h is nondecreasing; hence $h(x) \geq h(a) \geq 0$ for $x \geq a$. ∎

Example 1.2.2. To prove $e^x \geq 1 + x$ for $x \geq 0$: at $x = 0$ both sides equal 1, and $\frac{\mathrm{d}}{\mathrm{d}x}e^x = e^x \geq 1 = \frac{\mathrm{d}}{\mathrm{d}x}(1 + x)$ for $x \geq 0$. Done. Similarly $\sin x \leq x$ and $\ln(1 + x) \leq x$ for $x \geq 0$.

Remark 1.2.3. Proposition 1.2.1 is the baby case of a **comparison theorem**: one controls a solution of a differential *inequality* by a solution of a differential *equation*. The same idea, run at full strength, is how one proves that solutions of nonlinear equations stay bounded.

5. Nature loves to use derivatives

Almost every law of nature is stated as a rate, not as a value:

Newton's second law	$m\ddot{\mathbf{x}} = \mathbf{F}$
Radioactive decay	$\dot{N} = -\lambda N$
Newton's law of cooling	$\dot{u} = -k(u - T)$
Fourier's law of heat	$\mathbf{q} = -\kappa \boldsymbol{\nabla} u$
Fick's law of diffusion	$\mathbf{J} = -D \boldsymbol{\nabla} c$

Intuition . Why should this be? Because nature is *local*. A particle does not know where it will be in an hour, or where it was yesterday; it only knows the forces acting on it right now. A law of nature can therefore only be a statement about the present instant — and a statement about how a quantity is changing at the present instant is precisely a statement about its derivative.

Example 1.2.4 (A falling object)**.** An object of mass m falls in the atmosphere near sea level. Formulate a differential equation describing its motion.

Model. Let t denote time (seconds) and let $v = v(t)$ be the velocity ($\mathrm{m\,s^{-1}}$), taken *positive downwards*. Thus t is the independent variable and v is the dependent variable.

Newton's second law states that

$$F = ma = m\frac{\mathrm{d}v}{\mathrm{d}t},$$

where F is the net force on the object. Two forces act:

- gravity, of magnitude mg, acting downwards (positive), where $g \approx 9{,}8\,\mathrm{m\,s^{-2}}$;
- drag (air resistance), which we *assume* to be proportional to the velocity, of magnitude γv, acting upwards (negative). The constant $\gamma > 0$ is the **drag coefficient**, with units of $\mathrm{kg\,s^{-1}}$.

Hence $F = mg - \gamma v$, and the model is

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - \gamma v. \tag{1.2}$$

■

Note 1.2.5. Equation (1.2) contains three constants m, g, γ. We call m and γ **parameters**: they depend on the particular falling object and may range over many values during an experiment. By contrast g is a **physical constant**, the same for every object.

Taking $m = 10\,\mathrm{kg}$ and $\gamma = 2\,\mathrm{kg\,s^{-1}}$, equation (1.2) becomes

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 9.8 - \frac{v}{5}. \tag{1.3}$$

§1.2.ii. Direction Fields: Seeing the Dynamics Without Solving

Boyce §1.1

We do not yet know how to solve (1.3). The dynamic viewpoint says we should not need to: the equation already tells us the slope of the solution at every point of the plane.

Definition 1.2.6 (Direction field)**.** Consider a first-order equation in the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y), \tag{1.4}$$

where f — sometimes called the **rate function** — is a given function of the two variables t and y. The **direction field** (or **slope field**) of (1.4) is obtained by drawing, at each point (t, y) of a rectangular grid, a short line segment of slope $f(t, y)$.

Intuition . A solution of (1.4) is a function $y = y(t)$ whose graph is a curve in the ty-plane. At each of its points, that curve has slope $f(t, y)$. Hence **every line segment of the direction field is tangent to the solution curve through that point**: the direction field shows the "flow" that the solution curves must follow.

Remark 1.2.7. Two practical points deserve emphasis.

(i) Constructing a direction field requires *no* solution of (1.4); one only evaluates $f(t, y)$ many times. So direction fields can be drawn even for equations that are very hard — or impossible — to solve in closed form.

(ii) Repeated evaluation and plotting is exactly the kind of task a computer does well; a grid of a few hundred points usually suffices.

Example 1.2.8 (Behaviour of solutions of (1.3))**.** Read off from (1.3): if $v = 40$ then $\mathrm{d}v/\mathrm{d}t = 1.8 > 0$; if $v = 50$ then $\mathrm{d}v/\mathrm{d}t = -0.2 < 0$; if $v = 60$ then $\mathrm{d}v/\mathrm{d}t = -2.2 < 0$. So there is a critical value of v separating solutions that increase from those that decrease, found by setting $\mathrm{d}v/\mathrm{d}t = 0$:

$$9.8 - \frac{v}{5} = 0 \Leftrightarrow v = 5 \cdot 9.8 = 49\,\mathrm{m\,s^{-1}}.$$

Definition 1.2.9 (Equilibrium solution)**.** A constant solution $y(t) \equiv c$ of $\mathrm{d}y/\mathrm{d}t = f(t, y)$ is called an **equilibrium solution**. Equivalently, c is a root of $f(t, c) = 0$ for all t.

For (1.3) the equilibrium solution is $v(t) \equiv 49$: substituting it makes both sides zero. Physically it is the perfect balance between gravity and drag, and since all other solutions appear to converge to it, it is called the **terminal velocity**.

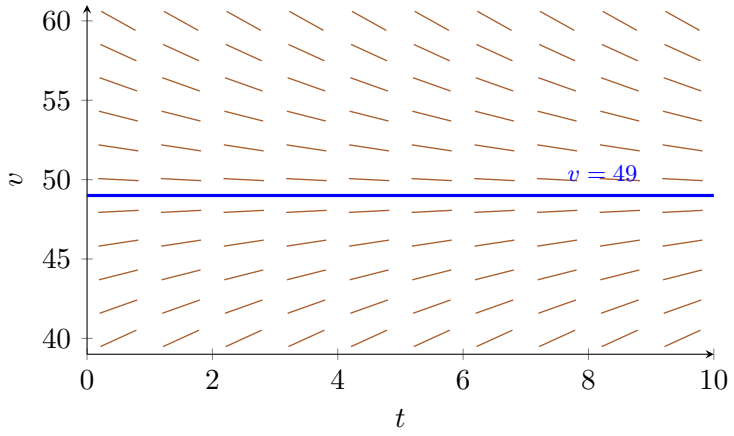


Figure 1.2.: Direction field and equilibrium solution of $\mathrm{d}v/\mathrm{d}t = 9.8 - v/5$. The equilibrium $v = 49$ *attracts* the other solutions.

§1.3. Examples

Boyce §1.1 (population, linear model); the remaining models are lecture material

§1.3.i. Population Growth

Example 1.3.1 (Field mice and owls)**.** A population of field mice inhabits a rural area. In the absence of predators we assume the population grows at a rate proportional to the current population: if $p(t)$ denotes the population at time t, then

$$\frac{\mathrm{d}p}{\mathrm{d}t} = rp, \tag{1.5}$$

where the proportionality factor r is the **rate constant** or **growth rate**. Suppose t is measured in months and $r = 0.5/\text{month}$.

Now suppose owls in the neighbourhood kill 15 field mice per *day*. Since time is measured in months, the predation term is -450 per month, and the model becomes

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{p}{2} - 450. \tag{1.6}$$

Setting $\mathrm{d}p/\mathrm{d}t = 0$ gives the equilibrium solution $p(t) \equiv 900$, at which growth and predation exactly balance.

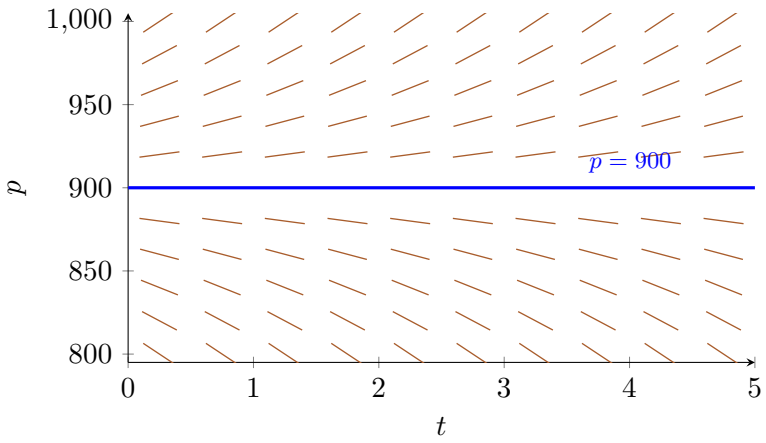


Figure 1.3.: Direction field and equilibrium solution of $\mathrm{d}p/\mathrm{d}t = p/2 - 450$. Here the equilibrium $p = 900$ *repels* the other solutions.

In both examples so far the equilibrium solution separates increasing solutions from decreasing ones. But the two behave oppositely: solutions of (1.3) are *attracted* by the equilibrium, while solutions of (1.6) are *repelled* by it. In the latter case the equilibrium itself will never be observed in practice — yet it is still the key to understanding the picture.

Remark 1.3.2 (Limitations)**.** Model (1.3) is valid only while the object falls freely; at large velocities the assumption that drag is *linear* in v must be replaced by a nonlinear one. Model (1.6) eventually predicts either a negative number of mice (if $p < 900$) or an absurdly large one (if $p > 900$), so it becomes unacceptable after a fairly short time. A better population model — the **logistic equation** — appears in Boyce §2.5.

§1.3.ii. Relaxation Processes and Equilibria

All the models below are the same equation wearing different clothes. We solve it once.

The linear model

Theorem 1.3.3 (General solution of $y' = ay - b$)
Let $a \neq 0$. The differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = ay - b \tag{1.7}$$

has general solution

$$y(t) = \frac{b}{a} + ce^{at}, \qquad c \in \mathbb{R} \text{ arbitrary}, \tag{1.8}$$

and the solution of the initial value problem $y' = ay - b$, $y(0) = y_0$ is

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a}\right) e^{at}. \tag{1.9}$$

The computation you would actually perform is to divide by $y - b/a$ and integrate a logarithm. That works, but it needs an argument to justify it (Remark 1.3.5 below), so we first give a proof that avoids division altogether.

Proof. **Step 1: every solution has the form (1.8).** Let y be *any* solution of (1.7) on an interval I, and define the auxiliary function

$$w(t) := \left(y(t) - \frac{b}{a}\right) e^{-at}, \qquad t \in I. \tag{1.10}$$

Differentiating with the product rule, and then using $y' = ay - b$,

$$w'(t) = y'(t)e^{-at} - a\left(y(t) - \frac{b}{a}\right) e^{-at} = e^{-at}\Big[\underbrace{y'(t)}_{=ay(t)-b} - ay(t) + b\Big] = e^{-at} \cdot 0 = 0$$

for every $t \in I$. A function with vanishing derivative on an *interval* is constant, so $w(t) = c$ for some $c \in \mathbb{R}$ and all $t \in I$. Solving (1.10) for y gives exactly (1.8).

Step 2: every function of the form (1.8) is a solution. If $y(t) = \frac{b}{a} + ce^{at}$, then $y'(t) = ac\,e^{at}$, while

$$ay(t) - b = a\left(\frac{b}{a} + ce^{at}\right) - b = \cancel{b} + ac\,e^{at} - \cancel{b} = ac\,e^{at} = y'(t). \checkmark$$

Step 3: the initial condition. Setting $t = 0$ in (1.8) gives $y_0 = \frac{b}{a} + c$, i.e. $c = y_0 - \frac{b}{a}$, which is (1.9). ∎

Remark 1.3.4. Step 1 never divides by anything and never takes a logarithm, so no value of y has to be excluded: the equilibrium solution $y \equiv b/a$ simply appears as the case $c = 0$, with no special pleading. The function e^{-at} in (1.10) is not magic — it is the **integrating factor** of Theorem 2.1.9, met here for the first time.

Remark 1.3.5 (What the logarithm computation really proves)**.** Here is the usual computation. Assuming $y(t) \neq b/a$, rewrite (1.7) as

$$\frac{\mathrm{d}y/\mathrm{d}t}{y - \frac{b}{a}} = a,$$

recognise the left-hand side as $\dfrac{\mathrm{d}}{\mathrm{d}t}\ln\left|y-\frac{b}{a}\right|$ by the chain rule, and integrate:

$$\ln\left|y(t)-\frac{b}{a}\right| = at + C \Rightarrow \left|y(t)-\frac{b}{a}\right| = e^{C}e^{at} \Rightarrow y(t)-\frac{b}{a} = \pm e^{C}e^{at}. \tag{1.11}$$

It gets the right answer, but as it stands it has *two* gaps.

(i) It divides by $y-b/a$, so it says nothing about a solution that takes the value b/a somewhere. The equilibrium solution has to be put back by hand, and since $e^{C} > 0$ the last step of (1.11) never produces $c = 0$ on its own.

(ii) **The sign in (1.11) is chosen *locally*.** Removing an absolute value is legitimate only on an interval where $y - b/a$ keeps a fixed sign. If y were above b/a for some times and below it for others, (1.11) would give $c = +e^{C}$ on some subintervals and $c = -e^{C}$ on others — and there would be no single constant c, so (1.8) would be false.

Gap (ii) is the serious one, and it is exactly the right thing to worry about. It closes because the bad case cannot occur:

Corollary 1.3.6 (A solution never crosses the equilibrium)

Let y be a solution of (1.7) on an interval I. Then *exactly one* of the following holds:

$$\text{(a) } y(t) = \frac{b}{a} \ \ \forall t \in I, \qquad \text{(b) } y(t) > \frac{b}{a} \ \ \forall t \in I, \qquad \text{(c) } y(t) < \frac{b}{a} \ \ \forall t \in I.$$

In particular $y - b/a$ has a constant sign, so the sign chosen in (1.11) is the *same* for all $t \in I$, and the logarithm computation is valid after all.

Proof. By Step 1 of the proof above, $y(t) - \frac{b}{a} = c\,e^{at}$ for a single constant c. Since $e^{at} > 0$ for every t, the sign of $y(t) - \frac{b}{a}$ equals the sign of c at every t: the three cases are $c = 0$, $c > 0$, $c < 0$. ∎

Intuition . Geometrically: the horizontal line $y = b/a$ is itself an integral curve, and Corollary 1.3.6 says no other integral curve ever touches it. It acts as a *barrier* dividing the plane into two halves, and a solution is trapped in whichever half it starts in. Look again at the direction fields on pages 7 and 8: above the equilibrium every segment points one way and below it every segment points the other, and nothing ever gets across.

This is a first instance of something general. Through each point of the plane there passes exactly *one* solution curve (Boyce ğ2.4), so two distinct solutions can never meet. A solution meeting the constant solution $y \equiv b/a$ at one instant would have to *be* it — which is case (a).

Example 1.3.7 (Concretely)**.** For the field mice, $\mathrm{d}p/\mathrm{d}t = \frac{p}{2} - 450$ has equilibrium $p = 900$ and general solution $p(t) = 900 + ce^{t/2}$. A population starting at $p_0 = 850$ has $c = -50 < 0$, so $p(t) = 900 - 50e^{t/2}$ stays strictly below 900 forever — it does not approach 900 from below and then overshoot; it runs away downwards. A population starting at $p_0 = 950$ has $c = +50$ and stays strictly above. No solution is ever on both sides.

Definition 1.3.8 (General solution, integral curves)**.** For $a \neq 0$, expression (1.8) contains *all* solutions of (1.7) and is called the **general solution**. Its geometric representation is an infinite family of curves called **integral curves**, one for each value of c. Satisfying an initial condition amounts to selecting the integral curve through the given initial point.

Exercise 1.3.9. Find the general solution of (1.7) in the remaining case $a = 0$. (Formula (1.8) does not apply.)

It is worth rewriting (1.9) in the form that makes the dynamics obvious. Put $y_{\text{eq}} = b/a$ and $k = -a$:

Corollary 1.3.10 (Relaxation form)

The initial value problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -k\,(y - y_{\text{eq}})\,, \qquad y(0) = y_0 \tag{1.12}$$

has the unique solution

$$y(t) = y_{\text{eq}} + (y_0 - y_{\text{eq}})\,e^{-kt}. \tag{1.13}$$

If $k > 0$, then $y(t) \to y_{\text{eq}}$ as $t \to \infty$: the equilibrium is **stable** (attracting). If $k < 0$ it is **unstable** (repelling).

Definition 1.3.11 (Relaxation time)**.** For $k > 0$, the quantity $\tau = 1/k$ is the **relaxation time** (or **time constant**) of (1.12): it is the time in which the deviation $y - y_{\text{eq}}$ shrinks by a factor of e. After 3τ about 95% of the initial deviation is gone, and after 5τ about 99%.

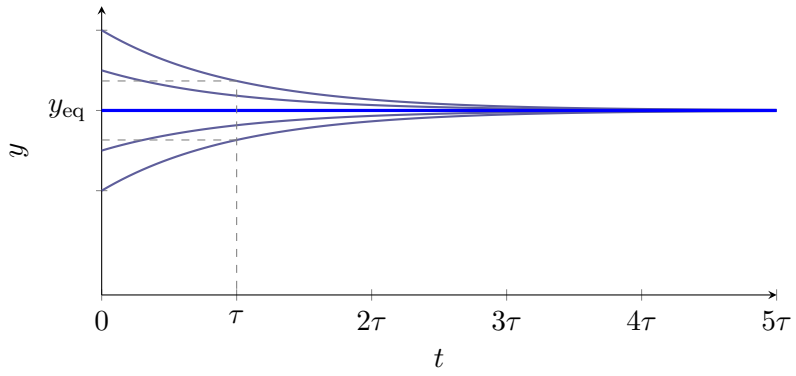


Figure 1.4.: Relaxation towards a stable equilibrium: every solution of (1.12) forgets its initial value exponentially, on the time scale $\tau = 1/k$.

Example 1.3.12 (The falling object, solved)**.** Equation (1.2) is (1.7) with $a = -\gamma/m$ and $b = -g$, so by (1.9)

$$v(t) = \frac{mg}{\gamma} + \left(v_0 - \frac{mg}{\gamma}\right) e^{-\gamma t/m}.$$

The terminal velocity is $v_{\text{eq}} = mg/\gamma$ and the relaxation time is $\tau = m/\gamma$: for a given mass, more drag means faster convergence.

Example 1.3.13 (The falling object, quantitatively)**.** Let $m = 10\,\text{kg}$, $\gamma = 2\,\text{kg}\,\text{s}^{-1}$, so that $\mathrm{d}v/\mathrm{d}t = 9.8 - v/5$, and drop the object ($v(0) = 0$) from a height of 300 m. How long does it take to reach the ground, and how fast is it then moving?

Solution. Here $v_{\text{eq}} = 49$ and $\tau = 5$, so by (1.13)

$$v(t) = 49\left(1 - e^{-t/5}\right). \tag{1.14}$$

Let $x(t)$ be the distance fallen, so $\mathrm{d}x/\mathrm{d}t = v$. Integrating (1.14) and using $x(0) = 0$,

$$x = 49t + 245e^{-t/5} - 245.$$

The impact time T satisfies $x(T) = 300$, i.e. $49T + 245e^{-T/5} - 245 = 300$. This transcendental equation has no elementary solution, but numerically $T \approx 10{,}51\,\mathrm{s}$, and then $v(T) \approx 43{,}01\,\mathrm{m\,s^{-1}}$ from (1.14). ■

Example 1.3.14 (Field mice, solved)**.** For $\mathrm{d}p/\mathrm{d}t = rp - k$ with $r, k > 0$, Theorem 1.3.3 gives

$$p(t) = \frac{k}{r} + \left(p_0 - \frac{k}{r}\right) e^{rt}.$$

Here $a = r > 0$, so the equilibrium k/r is *unstable*. If $p_0 = k/r$ the population is constant; if $p_0 > k/r$ it grows exponentially; if $p_0 < k/r$ it decreases and reaches zero *in finite time*, i.e. the mice go extinct. (Negative values of p, though allowed by the formula, are meaningless.)

Newton's law of cooling

Example 1.3.15 (Newton's law of cooling)**.** A body at temperature $u(t)$ sits in surroundings held at the ambient temperature T. Newton's law of cooling states that the body's temperature changes at a rate proportional to the difference between it and its surroundings:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = -k\,(u - T), \qquad k > 0. \tag{1.15}$$

By Corollary 1.3.10, $u(t) = T + (u_0 - T)e^{-kt}$.

Remark 1.3.16. Note that (1.15) is the same equation whether the body is hotter or cooler than its surroundings: the sign of $u - T$ takes care of itself. The constant k depends on the body and on how well it is insulated, not on the temperatures.

Example 1.3.17. Coffee at 95 °C is left in a room at 20 °C, and cools to 60 °C in 10 minutes. When will it be drinkable, say 50 °C?

Solution. Here $u(t) = 20 + 75e^{-kt}$. From $u(10) = 60$ we get $e^{-10k} = 40/75$, so $k = \frac{1}{10}\ln\frac{75}{40} \approx 0.0629$ per minute (a relaxation time $\tau \approx 16$ min). Then $u(t) = 50$ gives

$$t = \frac{1}{k}\ln\frac{75}{30} \approx 14.6 \text{ minutes}.$$

■

Reversible chemical reaction

Example 1.3.18 (A reversible reaction)**.** Consider the reversible reaction

$$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B,$$

in which A converts to B at rate k_1 and B converts back to A at rate k_2. Write $a(t)$ and $b(t)$ for the concentrations. Since no matter is created or destroyed, $a(t) + b(t) = c_0$ is constant, and

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -k_1 a + k_2 b = -k_1 a + k_2(c_0 - a) = -(k_1 + k_2)\left(a - \frac{k_2 c_0}{k_1 + k_2}\right).$$

Observation . This is (1.12) with

$$a_{\mathrm{eq}} = \frac{k_2 c_0}{k_1 + k_2}, \qquad b_{\mathrm{eq}} = \frac{k_1 c_0}{k_1 + k_2}, \qquad \tau = \frac{1}{k_1 + k_2}.$$

So the reaction relaxes to the equilibrium mixture at a rate governed by the *sum* of the

two rate constants, while the equilibrium composition is governed by their *ratio*:

$$K := \frac{b_{\text{eq}}}{a_{\text{eq}}} = \frac{k_1}{k_2},$$

which is exactly the **equilibrium constant** of the reaction. Chemical equilibrium is not the absence of reaction; it is the balance of two reactions that never stop.

Dynamic supply–demand model

Example 1.3.19 (A market that adjusts in time)**.** For a single good at price p, suppose the quantity demanded and the quantity supplied are

$$Q_d = \alpha - \beta p, \qquad Q_s = -\gamma + \delta p, \qquad \alpha, \beta, \gamma, \delta > 0.$$

The *static* (Walrasian) theory sets $Q_d = Q_s$ and solves for the market clearing price

$$p^* = \frac{\alpha + \gamma}{\beta + \delta}.$$

But nothing in that calculation says the market will ever *reach* p^*. The *dynamic* theory adds a rule of motion: the price rises when there is excess demand and falls when there is excess supply, at a rate proportional to the excess:

$$\frac{\mathrm{d}p}{\mathrm{d}t} = j\,(Q_d - Q_s)\,, \qquad j > 0. \tag{1.16}$$

Analysis. Substituting the two schedules into (1.16),

$$\frac{\mathrm{d}p}{\mathrm{d}t} = j\,(\alpha + \gamma - (\beta + \delta)p) = -j(\beta + \delta)\,(p - p^*)\,,$$

which is (1.12) with $k = j(\beta + \delta) > 0$. Hence

$$p(t) = p^* + (p_0 - p^*)\,e^{-j(\beta+\delta)t} \longrightarrow p^*.$$

■

The dynamic model answers a question the static one cannot even ask: the equilibrium price is not merely a solution of an equation, it is an *attractor*. Markets with stiff schedules (large $\beta + \delta$) or fast traders (large j) converge quickly; sluggish ones may take so long that the equilibrium is never observed.

Every model in subsection 1.3.ii — a falling body, a cooling cup of coffee, a reversible reaction, a market, and even the influence of your friends — is the *same* differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -k\,(y - y_{\text{eq}})\,.$$

Two numbers say everything: *where* the system settles (y_{eq}) and *how fast* it gets there ($\tau = 1/k$). This is what it means for mathematics to be useful: one calculation, many phenomena.

Note 1.3.20 (Historical background II: Euler, Lagrange, Laplace)**.** Leonhard Euler (1707–1783), the greatest mathematician of the eighteenth century, identified the condition for exactness of first-order equations and developed the theory of integrating factors (1734–35), gave the general solution of homogeneous linear equations with constant coefficients (1743), and proposed a numerical procedure (1768–69). Joseph-Louis Lagrange (1736–1813) showed (1762–65) that the general solution of a homogeneous nth order linear equation is a linear combination of n independent solutions, and developed variation of parameters (1774–75). Pierre-Simon de Laplace (1749–1827) gave his name to Laplace's equation and the Laplace transform.

§1.3.iii. Celestial Motion

Lecture material; not in Boyce

So far every example has been a single first-order equation with a single equilibrium. Celestial mechanics is where that comfortable picture ends.

The two-body problem: ellipse orbits

Example 1.3.21 (Two bodies under gravity)**.** Two bodies of masses m_1, m_2 at positions $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^3$ attract each other by Newton's law of gravitation:

$$m_1\ddot{\mathbf{x}}_1 = \frac{Gm_1m_2(\mathbf{x}_2-\mathbf{x}_1)}{\|\mathbf{x}_2-\mathbf{x}_1\|^3}, \qquad m_2\ddot{\mathbf{x}}_2 = \frac{Gm_1m_2(\mathbf{x}_1-\mathbf{x}_2)}{\|\mathbf{x}_1-\mathbf{x}_2\|^3}. \tag{1.17}$$

Remark 1.3.22. This is a *second-order system* of six scalar equations, equivalently a first-order system in the 12 unknowns (positions and velocities).

Writing $\mathbf{r} = \mathbf{x}_2 - \mathbf{x}_1$ for the relative position and $\mu = G(m_1+m_2)$, the two equations of (1.17) combine into one:

$$\ddot{\mathbf{r}} = -\mu\frac{\mathbf{r}}{\|\mathbf{r}\|^3}. \tag{1.18}$$

Theorem 1.3.23 (Kepler's laws; Newton 1687)
Every bounded solution of (1.18) satisfies:

(i) the orbit is an ellipse with the centre of mass at one focus;

(ii) the radius vector sweeps out equal areas in equal times;

(iii) the square of the period is proportional to the cube of the semi-major axis, $T^2 = \dfrac{4\pi^2}{\mu}a^3$.

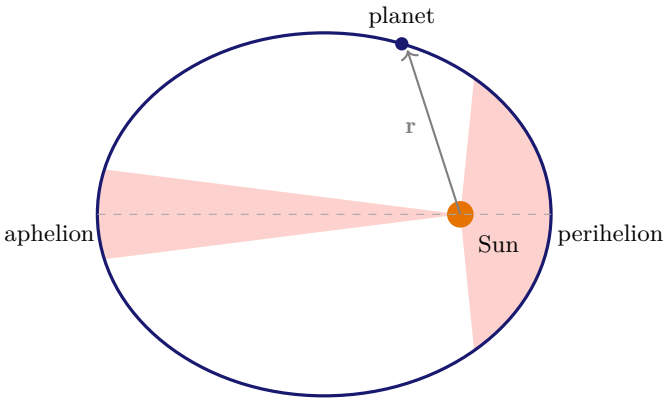


Figure 1.5.: The two-body problem: an elliptical orbit with the Sun at a focus. The two shaded regions have equal area and are swept out in equal times — so the planet moves fastest at perihelion.

The two-body problem is integrable: it has enough conserved quantities (energy, the three components of angular momentum, and the Laplace–Runge–Lenz vector) to be solved completely in closed form. It is one of the very few nonlinear systems for which this is true.

The three-body problem: the figure-eight orbit

Add one more body and everything breaks.

Fact 1.3.24. The three-body problem has no solution in closed form. Bruns (1887) and Poincaré (1890) showed that it admits no further algebraic conserved quantities of the kind that make the two-body problem solvable; Poincaré's work on it is the origin of the modern theory of chaos.

What one can do instead is look for special solutions of high symmetry. Euler (1767) found collinear ones and Lagrange (1772) found equilateral ones. In 1993 Cristopher Moore found numerically, and in 2000 Alain Chenciner and Richard Montgomery proved rigorously, that there is a solution in which three *equal* masses chase one another around a single figure-eight curve, each one-third of a period behind the last.

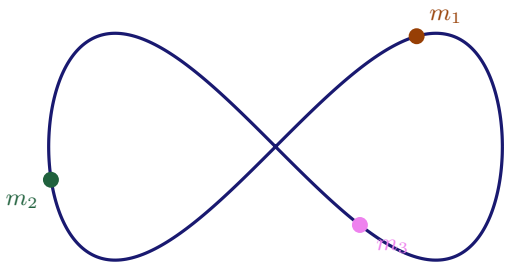


Figure 1.6.: The figure-eight orbit of the three-body problem (schematic, drawn as a lemniscate): three equal masses travel the *same* closed curve, spaced one third of a period apart. Discovered numerically by Moore (1993); existence proved by Chenciner and Montgomery (2000).

§1.3.iv. The Lorenz System and Chaos

Lecture material; Boyce treats this in §9.8

Example 1.3.25 (The Lorenz equations)**.** In 1963 Edward Lorenz, a meteorologist at MIT, reduced a model of convection in the atmosphere to three ordinary differential equations:

$$\begin{cases} \dot{x} = \sigma(y - x), \\ \dot{y} = x(\rho - z) - y, \\ \dot{z} = xy - \beta z, \end{cases} \qquad \sigma = 10, \quad \beta = \frac{8}{3}, \quad \rho = 28. \tag{1.19}$$

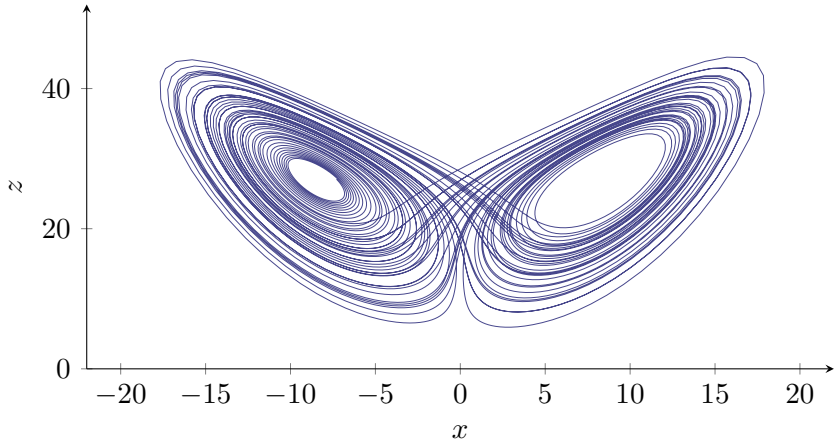


Figure 1.7.: A single trajectory of the Lorenz system (1.19), projected onto the xz-plane: the "butterfly" strange attractor. The curve never closes and never repeats, yet stays forever in a bounded region.

Observation . System (1.19) is completely *deterministic*: the state at time 0 determines the state at every later time. Nevertheless it is unpredictable in practice, because two initial conditions that differ by 10^{-6} end up on opposite wings of the butterfly within a few dozen time units. Lorenz discovered this by accident, when he restarted a simulation from three-digit printed values instead of the six-digit values in the machine and got a completely different forecast. This **sensitive dependence on initial conditions** is the defining feature of **chaos**, popularly the *butterfly effect*.

Remark 1.3.26 (Why three dimensions?)**.** Chaos cannot happen sooner. For a scalar autonomous equation $y' = f(y)$, solutions are monotone between equilibria, so they can only converge or blow up. In the plane, the Poincaré–Bendixson theorem says a bounded trajectory must approach an equilibrium or a closed orbit. Only from *three* dimensions on is there enough room for a trajectory to wander forever without repeating itself.

"Solvable in closed form" and "understood" are different things. Most of the equations that matter cannot be solved by the formulas of Chapter 2; the goal then shifts from finding $y(t)$ to describing the *qualitative* behaviour of the solutions — equilibria, stability, periodicity, attractors.

§1.4. Classification of Differential Equations

Boyce §1.3

Mastering the following vocabulary is essential for selecting appropriate solution methods.

§1.4.i. Ordinary and Partial Differential Equations

Definition 1.4.1 (ODE, PDE)**.** If the unknown function depends on a *single* independent variable, only ordinary derivatives appear and the equation is an **ordinary differential equation** (ODE). If it depends on *several* independent variables, the derivatives are partial derivatives and the equation is a **partial differential equation** (PDE).

Example 1.4.2. All equations so far have been ODEs. Another is the equation

$$L\frac{\mathrm{d}^2Q(t)}{\mathrm{d}t^2} + R\frac{\mathrm{d}Q(t)}{\mathrm{d}t} + \frac{1}{C}Q(t) = E(t)$$

for the charge $Q(t)$ on a capacitor in a circuit with capacitance C, resistance R and inductance L. Typical PDEs are the **heat conduction equation**

$$\alpha^2\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial u(x,t)}{\partial t}$$

and the **wave equation**

$$a^2\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial^2 u(x,t)}{\partial t^2},$$

in which u depends on the two independent variables x and t, and α^2, a^2 are physical constants.

§1.4.ii. Order

Definition 1.4.3 (Order)**.** The **order** of a differential equation is the order of the highest derivative appearing in it. The general nth order ODE is

$$F\left(t, u(t), u'(t), \ldots, u^{(n)}(t)\right) = 0. \tag{1.20}$$

> **Notation .** It is customary to write y for $u(t)$, and $y', y'', \ldots, y^{(n)}$ for $u'(t), u''(t), \ldots, u^{(n)}(t)$, so that (1.20) becomes
>
> $$F\left(t, y, y', \ldots, y^{(n)}\right) = 0.$$

Example 1.4.4. $y''' + 2e^t y'' + yy' = t^4$ is a third-order equation. The equations (1.17) of celestial mechanics are second order.

We will always assume that (1.20) can be solved for the highest derivative, giving the **explicit** (or **normal**) form

$$y^{(n)} = f\left(t, y, y', \ldots, y^{(n-1)}\right). \tag{1.21}$$

Remark 1.4.5 (Where the ambiguity comes from)**.** Passing from the implicit equation (1.20) to the explicit form (1.21) is not just rearranging terms: it means *solving* the equation

$$F\left(t, y, \ldots, y^{(n-1)}, y^{(n)}\right) = 0$$

for the last variable $y^{(n)}$, treating everything else as known. That is an algebraic step, and an algebraic equation can have more than one root. So a single implicit equation of the form (1.20) need not determine a single explicit equation of the form (1.21) — it may split into several, one per root.

Example 1.4.6. The equation $(y')^2 + ty' + 4y = 0$ is quadratic in the unknown y'. Applying the quadratic formula (with t and y playing the role of the coefficients),

$$y' = \frac{-t \pm \sqrt{t^2 - 16y}}{2},$$

which is really *two* different explicit equations,

$$y' = \frac{-t + \sqrt{t^2 - 16y}}{2} \qquad \text{and} \qquad y' = \frac{-t - \sqrt{t^2 - 16y}}{2},$$

each defined only where $t^2 \geq 16y$ (so the square root is real), and each with a *different* function $f(t, y)$ on the right-hand side. Either one, substituted back in, satisfies the original implicit equation — but the two are genuinely different differential equations, and in general have different solutions $y(t)$.

Remark 1.4.7. This is what "assume (1.20) can be solved for the highest derivative" really commits us to: pick *one* of the roots (here, one of the two square-root branches, on the region $t^2 \geq 16y$ where it is defined) and take that fixed choice as f in (1.21). Everything afterwards — in particular the definition of *solution* below — refers to that one explicit equation, not to the implicit equation we started from.

§1.4.iii. Scalar Equations and Systems of Equations

Definition 1.4.8 (System)**.** If a single unknown function is to be determined, one equation suffices and we speak of a **scalar equation**. If two or more unknown functions are involved, a **system** of differential equations is required.

Example 1.4.9 (Lotka–Volterra / predator–prey equations)**.**

$$\begin{cases} \dfrac{\mathrm{d}x}{\mathrm{d}t} = ax - \alpha xy, \\ \dfrac{\mathrm{d}y}{\mathrm{d}t} = -cy + \gamma xy, \end{cases}$$

where $x(t)$ and $y(t)$ are the populations of prey and predator respectively and $a, \alpha, c, \gamma > 0$ are empirical constants. The Lorenz system (1.19) is a system of three equations; the two-body problem (1.17) a system of six.

Remark 1.4.10. The two classifications interact: any scalar equation of order n can be rewritten as a *first-order system* of n equations, by taking $y_1 = y,\ y_2 = y',\ \ldots,\ y_n = y^{(n-1)}$. This is why the theory of first-order systems is the central object of the subject.

§1.4.iv. Linear and Nonlinear Equations

Definition 1.4.11 (Linear equation)**.** The ODE $F(t, y, y', \ldots, y^{(n)}) = 0$ is **linear** if F is a linear function of the variables $y, y', \ldots, y^{(n)}$. Thus the general linear ODE of order n is

$$a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \cdots + a_n(t)y = g(t). \tag{1.22}$$

An equation not of the form (1.22) is **nonlinear**.

Remark 1.4.12. Note that no condition is imposed on how t enters: the coefficients $a_i(t)$ and the right-hand side $g(t)$ may be arbitrary functions of t. Linearity concerns only y and its derivatives.

Example 1.4.13. $y''' + 2e^t y'' + yy' = t^4$ is nonlinear, because of the term yy'. Each Lotka–Volterra equation is nonlinear because of the products xy, and each Lorenz equation because of xz and xy.

Example 1.4.14 (The oscillating pendulum)**.** The angle $\theta = \theta(t)$ that a pendulum of length L makes with the vertical satisfies

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{L}\sin\theta = 0, \tag{1.23}$$

which is nonlinear because of the term $\sin\theta$.

Definition 1.4.15 (Linearization)**.** Approximating a nonlinear equation by a linear one is called **linearization**.

Example 1.4.16. For small θ we have $\sin\theta \approx \theta$, so (1.23) is approximated by the linear equation

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{L}\theta = 0.$$

The theory and methods for linear equations are highly developed; for nonlinear equations the theory is more complicated and the methods less satisfactory. Fortunately, many significant problems are linear or can be usefully linearized — but, as the Lorenz system shows, many phenomena genuinely require nonlinear equations.

§1.4.v. Solutions

Definition 1.4.17 (Solution)**.** A **solution** of the nth order ODE (1.21) on an interval $\alpha < t < \beta$ is a function φ such that $\varphi', \varphi'', \dots, \varphi^{(n)}$ exist and satisfy

$$\varphi^{(n)}(t) = f\left(t, \varphi(t), \varphi'(t), \dots, \varphi^{(n-1)}(t)\right)$$

for every t with $\alpha < t < \beta$.

Unless stated otherwise we take f real-valued and seek real-valued solutions.

Observation . Finding a solution can be very hard, but *checking* a candidate is usually easy: just substitute it into the equation. For instance, $y_1(t) = \cos t$ satisfies $y'' + y = 0$ for all t, since $y_1'(t) = -\sin t$ and $y_1''(t) = -\cos t$; likewise $y_2(t) = \sin t$ is a solution. Verifying a proposed solution is a habit worth acquiring.

§1.4.vi. Three Fundamental Questions

Question 1.4.18 (Existence)**.** Does an equation of the form (1.21) always have a solution?

The answer is **no**: merely writing down such an equation does not guarantee a function satisfying it. Existence theorems state that under suitable restrictions on f, solutions do exist. This is not a purely theoretical concern: we would rather know in advance than waste effort on an unsolvable problem, and if a sensible physical problem yields an equation with no solution, something is wrong with the formulation.

Question 1.4.19 (Uniqueness)**.** If solutions exist, how many are there, and what extra conditions single one out?

Solutions generally contain arbitrary constants of integration, so there is typically an infinite family. Prescribing y at a point determines the constant — but this does not by itself rule out *other* solutions with the same prescribed value. If we know the problem has a unique solution and we have found one, we are finished; otherwise we should keep searching.

Question 1.4.20 (Determination of solutions)**.** Given (1.21), can we actually determine a solution, and if so, how?

Finding a solution simultaneously answers the existence question. But without existence theory one might, for example, use a computer to produce a numerical approximation to a "solution" that does not exist. Moreover, even when a solution exists it may not be expressible in terms of elementary functions — polynomial, trigonometric, exponential, logarithmic and hyperbolic — and unfortunately this is the situation for *most* differential equations. Hence we study both exact elementary methods for simple problems and approximation methods of a more general nature.

Always try to combine numerical, graphical and analytical methods, so as to attain maximum understanding of the behaviour of the solution and of the process it models.

2. First-Order Differential Equations

This chapter deals with equations of first order

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y),$$

where f is a given function of two variables. For an arbitrary f there is *no* general method of solution in terms of elementary functions; instead we describe several methods, each applicable to a certain subclass. The most important are **linear** equations (Section 2.1), **separable** equations (Section 2.2), and **exact** equations (Boyce §2.6).

§2.1. Linear Differential Equations; Method of Integrating Factors

Boyce §2.1

Definition 2.1.1 (First-order linear equation)**.** If f depends linearly on y, then $\mathrm{d}y/\mathrm{d}t = f(t, y)$ is a **first-order linear differential equation**. We write it in the **standard form**

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)y = g(t), \tag{2.1}$$

where p and g are given functions of t.

Remark 2.1.2. Sometimes it is convenient to write the equation as $P(t)y' + Q(t)y = G(t)$; as long as $P(t) \neq 0$, dividing by $P(t)$ returns it to the standard form (2.1). **Always convert to standard form before reading off p and g.**

Remark 2.1.3. Theorem 1.3.3 solved the special case p, g constant. We now allow both to depend on t, which no longer yields to the trick used there.

Occasionally the equation can be integrated at once.

Example 2.1.4. Solve $(4 + t^2)\dfrac{\mathrm{d}y}{\mathrm{d}t} + 2ty = 4t$.

Solution. The left-hand side is exactly the derivative of a product:

$$(4 + t^2)\frac{\mathrm{d}y}{\mathrm{d}t} + 2ty = \frac{\mathrm{d}}{\mathrm{d}t}\left[(4 + t^2)y\right].$$

So the equation reads $\dfrac{\mathrm{d}}{\mathrm{d}t}\left[(4 + t^2)y\right] = 4t$, and even though y is unknown we may integrate both sides:

$$(4 + t^2)y = 2t^2 + c \Rightarrow y = \frac{2t^2}{4 + t^2} + \frac{c}{4 + t^2}.$$

■

Most equations are not so lucky: the left-hand side is usually not the derivative of a product. Leibniz's idea was to *make* it one.

Definition 2.1.5 (Integrating factor)**.** A function $\mu(t)$ such that multiplying (2.1) by $\mu(t)$ turns the left-hand side into $\dfrac{\mathrm{d}}{\mathrm{d}t}\left[\mu(t)y\right]$ is called an **integrating factor** for (2.1).

Example 2.1.6. Find the general solution of $\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{2}y = \frac{1}{2}e^{t/3}$, and the particular solution through $(0, 1)$.

Solution. Multiply by an as-yet-undetermined $\mu(t)$:

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{2}\mu(t)y = \frac{1}{2}\mu(t)e^{t/3}. \tag{2.2}$$

For any differentiable function μ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mu(t)y\right) = \mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\mathrm{d}\mu(t)}{\mathrm{d}t}y,$$

so the left-hand side of (2.2) is such a derivative precisely when

$$\frac{\mathrm{d}\mu(t)}{\mathrm{d}t} = \frac{1}{2}\mu(t) \Leftrightarrow \frac{1}{\mu(t)}\frac{\mathrm{d}\mu(t)}{\mathrm{d}t} = \frac{1}{2} \Leftrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\ln|\mu(t)| = \frac{1}{2},$$

whence $\mu(t) = c_0 e^{t/2}$ for an arbitrary constant c_0. We do not need the most general integrating factor, so take $c_0 = 1$ and $\mu(t) = e^{t/2}$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{t/2}y\right) = \frac{1}{2}e^{5t/6} \Rightarrow e^{t/2}y = \frac{3}{5}e^{5t/6} + c,$$

and therefore the general solution is

$$y = \frac{3}{5}e^{t/3} + ce^{-t/2}.$$

Setting $t = 0$, $y = 1$ gives $1 = 3/5 + c$, so $c = 2/5$ and the desired solution is $y = \frac{3}{5}e^{t/3} + \frac{2}{5}e^{-t/2}$. ■

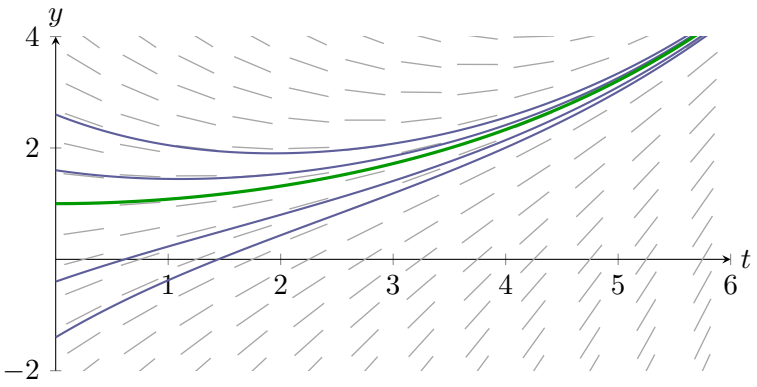


Figure 2.1.: Direction field and integral curves of $y' + \frac{1}{2}y = \frac{1}{2}e^{t/3}$; the green curve is the solution through $(0, 1)$.

§2.1.i. The Constant-Coefficient Case

For $\frac{\mathrm{d}y}{\mathrm{d}t} + ay = g(t)$ with a constant, the integrating factor must satisfy $\mathrm{d}\mu/\mathrm{d}t = a\mu$, so $\mu(t) = e^{at}$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{at}y\right) = e^{at}g(t) \Rightarrow e^{at}y = \int e^{at}g(t)\mathrm{d}t + c.$$

For complicated g the integral may have to be left unevaluated, in which case

$$y = e^{-at}\int_{t_0}^{t} e^{as}g(s)\mathrm{d}s + ce^{-at}. \tag{2.3}$$

Note 2.1.7. In (2.3) we use s for the integration variable to distinguish it from the independent variable t, and t_0 is any convenient lower limit. The choice of t_0 changes the value of c — indeed $c = y(t_0)e^{at_0}$ — but not the solution itself.

Example 2.1.8. Find the general solution of $\dfrac{\mathrm{d}y}{\mathrm{d}t} - 2y = 4 - t$ and discuss the behaviour as $t \to \infty$.

Solution. Here $a = -2$, so $\mu(t) = e^{-2t}$ and

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-2t}y\right) = 4e^{-2t} - te^{-2t}.$$

Integrating (by parts on the last term),

$$e^{-2t}y = -2e^{-2t} + \frac{1}{2}te^{-2t} + \frac{1}{4}e^{-2t} + c,$$

so the general solution is

$$y = -\frac{7}{4} + \frac{1}{2}t + ce^{2t}.$$

The behaviour for large t is dominated by ce^{2t}. If $c \neq 0$ the solution grows exponentially with the sign of c; all such solutions diverge. The borderline case is $c = 0$, i.e. $y(0) = -7/4$, for which $y = -\frac{7}{4} + \frac{1}{2}t$ grows only *linearly*. ■

§2.1.ii. The General Case

Theorem 2.1.9 (Method of integrating factors)
Consider the first-order linear equation in standard form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)y = g(t).$$

Then

$$\mu(t) = \exp\left(\int p(t)\mathrm{d}t\right) \tag{2.4}$$

is an integrating factor, and the general solution is

$$y = \frac{1}{\mu(t)}\left[\int_{t_0}^{t} \mu(s)g(s)\mathrm{d}s + c\right]. \tag{2.5}$$

Proof. Multiplying the equation by $\mu(t)$ gives

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)\mu(t)y = \mu(t)g(t), \tag{2.6}$$

whose left-hand side is $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mu(t)y)$ precisely when

$$\frac{\mathrm{d}\mu(t)}{\mathrm{d}t} = p(t)\mu(t).$$

Assuming temporarily that $\mu(t) > 0$, this says $\dfrac{1}{\mu(t)}\dfrac{\mathrm{d}\mu(t)}{\mathrm{d}t} = p(t)$, so $\ln|\mu(t)| = \int p(t)\mathrm{d}t + k$. Choosing $k = 0$ gives the simplest choice (2.4), which is indeed positive for all t, as assumed. Then equation (2.6) reads $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mu(t)y) = \mu(t)g(t)$, and integrating gives $\mu(t)y = \int \mu(t)g(t)\mathrm{d}t + c$, which is (2.5). ■

Remark 2.1.10. Observe that (2.5) involves **two** integrations: one to obtain $\mu(t)$ from (2.4), and another to obtain y.

Example 2.1.11. Solve the initial value problem $ty' + 2y = 4t^2$, $y(1) = 2$.

Solution. First convert to standard form (dividing by t):

$$y' + \frac{2}{t}y = 4t,$$

so $p(t) = 2/t$ and $g(t) = 4t$. The integrating factor is

$$\mu(t) = \exp\left(\int \frac{2}{t}\mathrm{d}t\right) = e^{2\ln|t|} = t^2.$$

Multiplying through, $t^2y' + 2ty = (t^2y)' = 4t^3$, so $t^2y = \int 4t^3\mathrm{d}t = t^4 + c$, and for $t > 0$,

$$y = t^2 + \frac{c}{t^2}.$$

The initial condition gives $2 = 1 + c$, so $c = 1$ and

$$y = t^2 + \frac{1}{t^2}, \qquad t > 0.$$

■

Remark 2.1.12 (Interval of validity)**.** This is the first example in which the solution fails to exist for all t: the solution becomes unbounded as $t \to 0^+$, so it is restricted to $0 < t < \infty$. It is worth being precise about *why* $t < 0$ is excluded, since it is **not** for the same reason as Remark 1.3.5 — that one is about an algebraic equation for y' having several roots; nothing like that is happening here.

First, check directly that $t < 0$ is not an algebra problem: substituting $y = t^2 + 1/t^2$ into the *original* equation $ty' + 2y = 4t^2$ gives

$$t\left(2t - \frac{2}{t^3}\right) + 2\left(t^2 + \frac{1}{t^2}\right) = 2t^2 - \frac{2}{t^2} + 2t^2 + \frac{2}{t^2} = 4t^2,$$

which holds for *every* $t \neq 0$, negative t included — there is no hidden sign restriction, no square root, nothing multivalued. So the formula itself is perfectly happy on $t < 0$.

The real obstruction is *connectedness*. Dividing by t to reach standard form is only legal for $t \neq 0$, and $p(t) = 2/t$ genuinely blows up there; so the domain on which our equation makes sense at all, $t \neq 0$, is not one interval but *two*: $(-\infty, 0)$ and $(0, \infty)$, with no way to pass between them. Every integral curve $y = t^2 + c/t^2$ with $c \neq 0$ shares this feature — it blows up as $t \to 0$ from whichever side it lives on — so no solution curve can be continued across $t = 0$ to link the two sides into a single differentiable function. But Definition 1.4.17 *requires* a solution to be one function on one interval $\alpha < t < \beta$. Since the initial point $t_0 = 1$ lies in $(0, \infty)$, that is the only interval available to the solution of *this* IVP, and $t < 0$ is simply outside its domain — not because the formula fails there, but because that region belongs to a different, disconnected piece of the picture.

This also means the constant on the two sides need not match. Because $y = t^2 + c/t^2$ is an *even* function of t for every fixed c, the left half of the plot below happens to look like a mirror image of the right half — but that symmetry is an accident of this particular formula, not a consequence of the initial condition. Nothing forces the branch on $t < 0$ to use $c = 1$; any other constant would glue on there just as validly, since the initial condition $y(1) = 2$ says nothing at all about behaviour on $t < 0$.

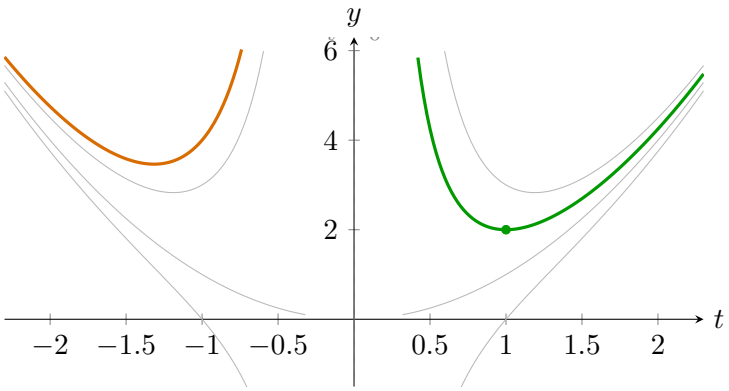


Figure 2.2.: No integral curve of $ty' + 2y = 4t^2$ crosses $t = 0$ (dashed): every curve with $c \neq 0$ blows up there. The green curve, $c = 1$, is the solution of the IVP $y(1) = 2$; it lives only on $t > 0$. The formula happens to be even in t, so the same shapes reappear for $t < 0$ (thin gray) — but the initial condition never touches that side, so the orange curve ($c = 3$) would glue on just as legitimately as any other choice.

Observation (The one exception)**.** With $y(1) = y_0$ instead of $y(1) = 2$, the same computation gives $c = y_0 - 1$, so $y = t^2 + (y_0 - 1)/t^2$. Every value of c makes the curve blow up at $t = 0$ — *except* $c = 0$, i.e. $y_0 = 1$, for which the singular term vanishes and $y = t^2$ is perfectly smooth at $t = 0$ (indeed $y = t^2$ solves $ty' + 2y = 4t^2$ for *all* $t \in \mathbb{R}$, no restriction needed). This is the one curve in the whole family that manages to cross from $t > 0$ to $t < 0$ without a jump. Solutions with $c > 0$ shoot up to $+\infty$ as $t \to 0^+$ (asymptotic to the positive y-axis) and those with $c < 0$ to $-\infty$ (the negative y-axis) — exactly what the thin gray curves in the figure show.

Example 2.1.13. Solve the initial value problem $2y' + ty = 2$, $y(0) = 1$.

Solution. In standard form, $y' + \frac{t}{2}y = 1$, so $p(t) = t/2$ and $\mu(t) = \exp\left(t^2/4\right)$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{t^2/4}y\right) = e^{t^2/4} \Rightarrow e^{t^2/4}y = \int e^{t^2/4}\mathrm{d}t + c.$$

The integral $\int e^{t^2/4}\mathrm{d}t$ **cannot** be evaluated in terms of elementary functions, so we leave it. Choosing the initial point $t = 0$ as the lower limit,

$$y = e^{-t^2/4}\int_0^t e^{s^2/4}\mathrm{d}s + ce^{-t^2/4}.$$

Setting $t = 0$, $y = 1$ gives $1 = 0 + c$, so $c = 1$. ■

Sometimes the solution must be left in terms of an integral. This is at most a slight inconvenience rather than a serious obstacle: for each t the integral is a *definite* integral, which can be approximated numerically to any desired accuracy.

Lecture 2

9/14–9/17

§2.2. Separable Differential Equations

Boyce §2.2

The direct-integration idea of Theorem 1.3.3 applies to a much larger class than the linear equations — including many nonlinear ones.

> **Notation .** In this section we use x rather than t for the independent variable, both because different letters are frequently used and because t will be reserved for another purpose (see Note 2.2.9 below).

Starting from the general first-order equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = f(x,y)$, rewrite it as

$$M(x,y) + N(x,y)\frac{\mathrm{d}y}{\mathrm{d}x} = 0. \tag{2.7}$$

This is always possible — take $M = -f$ and $N = 1$ — but there may be other ways.

Definition 2.2.1 (Separable equation)**.** If in (2.7) M is a function of x only and N is a function of y only, i.e.

$$M(x) + N(y)\frac{\mathrm{d}y}{\mathrm{d}x} = 0, \tag{2.8}$$

then the equation is called **separable**: writing it in the **differential form**

$$M(x)\mathrm{d}x + N(y)\mathrm{d}y = 0 \tag{2.9}$$

the terms involving each variable may be placed on opposite sides.

Remark 2.2.2. The differential form (2.9) is more symmetric and tends to suppress the distinction between the independent and dependent variables.

§2.2.i. Examples

Example 2.2.3. Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2}{1-y^2}$ is separable and find an equation for its integral curves.

Solution. Write it as $-x^2 + (1-y^2)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$, which has the form (2.8). Recall the chain rule: if y is a function of x, then

$$\frac{\mathrm{d}}{\mathrm{d}x}F(y) = \frac{\mathrm{d}}{\mathrm{d}y}F(y)\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = F'(y)\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Taking $F(y) = y - y^3/3$ gives $F'(y) = 1-y^2$, so the second term is $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(y - \frac{y^3}{3}\right)$; the first term is $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{x^3}{3}\right)$. Hence the equation reads

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{x^3}{3} + y - \frac{y^3}{3}\right) = 0,$$

and integrating (and multiplying by 3),

$$-x^3 + 3y - y^3 = c.$$

Any differentiable $y = \varphi(x)$ satisfying this is a solution. The integral curve through a given point (x_0, y_0) is found by substituting to determine c. ■

The same argument works in general.

Theorem 2.2.4 (Solution of a separable equation)

Let H_1, H_2 be any antiderivatives of M, N, so that $H_1' = M$ and $H_2' = N$. Then every differentiable $y = \varphi(x)$ satisfying

$$H_1(x) + H_2(y) = c \tag{2.10}$$

is a solution of (2.8). Moreover, the solution of the initial value problem $M(x)+N(y)y' = 0$, $y(x_0) = y_0$ is given implicitly by

$$\int_{x_0}^{x} M(s)\mathrm{d}s + \int_{y_0}^{y} N(s)\mathrm{d}s = 0. \tag{2.11}$$

Proof. Equation (2.8) becomes $H_1'(x)+H_2'(y)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$. Regarding y as a function of x, the chain rule gives

$$H_2'(y)\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}y}H_2(y)\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}H_2(y),$$

so the equation is $\dfrac{\mathrm{d}}{\mathrm{d}x}\left[H_1(x) + H_2(y)\right] = 0$. Integrating with respect to x gives (2.10).

For the initial value problem, setting $x = x_0$, $y = y_0$ in (2.10) gives $c = H_1(x_0) + H_2(y_0)$. Substituting this back and using

$$H_1(x) - H_1(x_0) = \int_{x_0}^{x} M(s)\mathrm{d}s, \qquad H_2(y) - H_2(y_0) = \int_{y_0}^{y} N(s)\mathrm{d}s$$

gives (2.11). ∎

Remark 2.2.5. Equation (2.10) defines the solution *implicitly* rather than explicitly. In practice one obtains it from (2.9) by integrating the first term with respect to x and the second with respect to y — the justification is exactly the argument above. To get an explicit formula one must then solve for y, which is often impossible analytically.

Example 2.2.6. Solve $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{3x^2+4x+2}{2(y-1)}$, $y(0) = -1$, and determine the interval on which the solution exists.

Solution. In differential form, $2(y-1)\mathrm{d}y = (3x^2+4x+2)\mathrm{d}x$. Integrating each side,

$$y^2 - 2y = x^3 + 2x^2 + 2x + c.$$

Substituting $x = 0$, $y = -1$ gives $c = 3$, so implicitly

$$y^2 - 2y = x^3 + 2x^2 + 2x + 3.$$

This is quadratic in y, so we can solve explicitly:

$$y = 1 \pm \sqrt{x^3 + 2x^2 + 2x + 4}.$$

Only *one* of these two branches satisfies the initial condition — the minus sign — so

$$y = \varphi(x) = 1 - \sqrt{x^3 + 2x^2 + 2x + 4}.$$

(The plus sign would give the solution with $y(0) = 3$ instead.) The solution is valid where the radicand is positive; its only real zero is $x = -2$, so the interval of validity is $x > -2$. Note that the boundary of that interval is the point $(-2, 1)$, at which the tangent line is *vertical*. ∎

Example 2.2.7. Solve $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{4x - x^3}{4 + y^3}$, find the solution through $(0, 1)$, and determine its interval of validity.

Solution. Rewriting as $(4+y^3)\mathrm{d}y = (4x-x^3)\mathrm{d}x$, integrating, multiplying by 4 and rearranging,

$$y^4 + 16y + x^4 - 8x^2 = c.$$

Substituting $x = 0$, $y = 1$ gives $c = 17$, so the solution is given implicitly by

$$y^4 + 16y + x^4 - 8x^2 = 17.$$

Here we *cannot* solve explicitly for y. The interval of validity extends on either side of the initial point as long as the function remains differentiable, i.e. until the tangent line becomes vertical. From the differential equation, this happens where $4 + y^3 = 0$, i.e. $y = (-4)^{1/3} \approx -1.5874$; the corresponding values are $x \approx \pm 3.3488$. ■

Note 2.2.8 (Constant solutions)**.** If $f(x, y_0) = 0$ for some value y_0 and *all* x, then the constant function $y \equiv y_0$ is a solution of $\mathrm{d}y/\mathrm{d}x = f(x, y)$. Such solutions are usually easy to spot. For example

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(y-3)\cos x}{1 + 2y^2}$$

has the constant solution $y \equiv 3$; the other solutions are found by separating variables.

Note 2.2.9 (Introducing a parameter)**.** It sometimes helps to regard *both* x and y as functions of a third variable t, so that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}.$$

Comparing numerators and denominators, the equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{F(x,y)}{G(x,y)}$ becomes the system

$$\frac{\mathrm{d}x}{\mathrm{d}t} = G(x, y), \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = F(x, y).$$

Replacing one equation by a pair may seem like a step backwards, but the system is often more amenable to investigation.

Note 2.2.10 (On the phrase "solve the equation")**.** Being able to solve explicitly for y, as in Example 2.2.6, is exceptional; often it is better to leave the solution in implicit form. Accordingly, "solve the following differential equation" means: find the solution explicitly if convenient, and otherwise find an equation defining it implicitly.

§2.2.ii. Homogeneous Equations $y' = g(y/x)$

Boyce §2.2, Problem 25 — promoted to a method here

Some equations that are *not* separable become separable after a change of the dependent variable. The most important such family is the following.

Definition 2.2.11 (Homogeneous equation)**.** The equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = f(x, y)$ is called **homogeneous** if the right-hand side depends on x and y only through their ratio, i.e. if there is a function g of one variable with

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g\left(\frac{y}{x}\right). \tag{2.12}$$

Abuse of Notation 2.2.12. The word "homogeneous" has a completely different meaning for linear equations (there it means $g(t) \equiv 0$ in (2.1), as in Boyce Chapter 3). The two notions have nothing to do with each other; only the context distinguishes them.

Observation (Why the name, and what it looks like)**.** Condition (2.12) says precisely that f is invariant under the scaling $(x, y) \mapsto (\lambda x, \lambda y)$:

$$f(\lambda x, \lambda y) = f(x, y) \quad \text{for all } \lambda \neq 0.$$

Geometrically, **the direction field is constant along every ray through the origin**. So the whole picture is determined by what happens on a single circle around the origin, and it should not be a surprise that one variable can be eliminated.

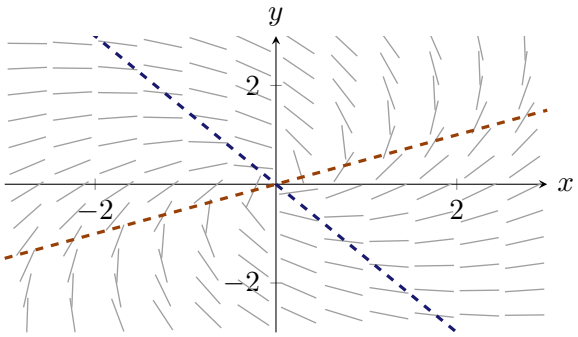


Figure 2.3.: Direction field of the homogeneous equation $y' = (x+y)/(x-y)$. Along each dashed ray $y = vx$ the slope is the same constant $g(v)$ — this is what "homogeneous" means.

Theorem 2.2.13 (Reduction to a separable equation)

Substituting

$$v = \frac{y}{x}, \qquad \text{i.e.} \qquad y = vx,$$

turns the homogeneous equation (2.12) into the separable equation

$$x\frac{\mathrm{d}v}{\mathrm{d}x} = g(v) - v. \tag{2.13}$$

Proof. Regard v as a new unknown function of x. By the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}(vx) = v + x\frac{\mathrm{d}v}{\mathrm{d}x},$$

while the right-hand side of (2.12) is $g(v)$. Equating the two and rearranging gives (2.13), in which the variables separate as

$$\frac{\mathrm{d}v}{g(v) - v} = \frac{\mathrm{d}x}{x} \qquad \text{whenever } g(v) \neq v. \qquad \blacksquare$$

Remark 2.2.14 (Equilibria of the reduced equation)**.** The roots of $g(v) = v$ — excluded above — are exactly the equilibrium solutions of (2.13). Each root v_0 gives a *straight-line* solution $y = v_0 x$ of the original equation. These are worth finding first; they are the rays along which the direction field points radially.

Example 2.2.15. Solve $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2 + xy + y^2}{x^2}$.

Solution. Dividing numerator and denominator by x^2,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2 = g\left(\frac{y}{x}\right), \qquad g(v) = 1 + v + v^2,$$

so the equation is homogeneous. Note $g(v) - v = 1 + v^2$, which never vanishes, so there are no straight-line solutions. By Theorem 2.2.13,

$$x\frac{\mathrm{d}v}{\mathrm{d}x} = 1 + v^2 \Rightarrow \frac{\mathrm{d}v}{1+v^2} = \frac{\mathrm{d}x}{x} \Rightarrow \arctan v = \ln|x| + c.$$

Hence $v = \tan(\ln|x| + c)$, and restoring $y = vx$,

$$y = x\tan(\ln|x| + c).$$

■

Example 2.2.16. Solve $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y^2 + 2xy}{x^2}$.

Solution. Here $g(v) = v^2 + 2v$, so $g(v) - v = v^2 + v = v(v+1)$. The roots $v = 0$ and $v = -1$ give the straight-line solutions $y = 0$ and $y = -x$. Otherwise,

$$\frac{\mathrm{d}v}{v(v+1)} = \frac{\mathrm{d}x}{x} \Rightarrow \int\left(\frac{1}{v} - \frac{1}{v+1}\right)\mathrm{d}v = \ln|x| + c,$$

using partial fractions. Thus $\ln\left|\dfrac{v}{v+1}\right| = \ln|x| + c$, i.e. $\dfrac{v}{v+1} = Kx$ for a constant K. Substituting $v = y/x$ and solving,

$$\frac{y}{y+x} = Kx \Rightarrow y = \frac{Kx^2}{1-Kx}.$$

■

The pattern is worth remembering, because it recurs: *find a change of variable that turns a problem you cannot do into one you can.* Separable equations are the base case; "homogeneous" is the first of many families that reduce to it.

§2.3. Modeling with First-Order Differential Equations

Boyce §2.3

Regardless of the field of application, three steps are always present in mathematical modelling.

Step 1: Construction of the model. Translate the physical situation into mathematical terms, stating clearly the physical principle(s) believed to govern the process. Each such principle involves a *rate of change* and hence leads to a differential equation.

Step 2: Analysis of the model. Solve the resulting equation(s) or, failing that, find out as much as possible about the properties of the solution. Further approximations may be needed here (linearizing a nonlinear equation, replacing a slowly varying coefficient by a constant) — but each must be checked against the physics.

Step 3: Comparison with experiment or observation. Interpret the solution in the context in which the problem arose, and check that it is physically reasonable. Serious inconsistency with observation means either an error in the solution, a model in need of refinement, or observations needing more care.

Remark 2.3.1. The equations are almost always only an *approximate* description of the actual process: bodies moving near the speed of light are not governed by Newton's laws, insect populations do not grow indefinitely, heat transfer depends on more than the temperature difference. Modelling also often replaces a discrete process by a continuous one — an insect population changes by whole numbers, but if it is large it is reasonable to treat it as a continuous variable and even to differentiate it.

§2.3.i. Mixing

Example 2.3.2 (A salt tank)**.** At time $t = 0$ a tank contains Q_0 lb of salt dissolved in 100 gal of water. Water containing $\frac{1}{4}$ lb of salt per gallon enters at r gal/min, and the well-stirred mixture drains at the same rate. Find $Q(t)$, the amount of salt at time t, and the limiting amount Q_L. If $r = 3$ and $Q_0 = 2Q_L$, find the time T after which the salt level is within 2% of Q_L; and find the flow rate needed so that $T \le 45$ min.

Solution. Salt is neither created nor destroyed, so

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \text{rate in} - \text{rate out}. \tag{2.14}$$

- **Rate in:** concentration $\times$ flow rate $= \frac{1}{4} \cdot r = r/4$ lb/min.
- **Rate out:** since inflow and outflow rates are equal, the volume stays at 100 gal; since the mixture is well stirred, the concentration is uniform and equal to $Q(t)/100$ lb/gal. So the rate out is $rQ(t)/100$ lb/min.

Hence the IVP is

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{r}{4} - \frac{rQ}{100}, \qquad Q(0) = Q_0. \tag{2.15}$$

Setting $\mathrm{d}Q/\mathrm{d}t = 0$ gives $Q_L = 25$ lb immediately — as physical intuition suggests, since the tank is eventually filled with the incoming mixture of concentration $\frac{1}{4}$ lb/gal.

Equation (2.15) is linear (and also separable). In standard form, $Q' + \frac{r}{100}Q = \frac{r}{4}$, with integrating factor $e^{rt/100}$, giving the general solution $Q(t) = 25 + ce^{-rt/100}$. The initial condition gives $c = Q_0 - 25$, so

$$Q(t) = 25 + (Q_0 - 25)e^{-rt/100} = \underbrace{25\left(1 - e^{-rt/100}\right)}_{\text{salt from the inflow}} + \underbrace{Q_0 e^{-rt/100}}_{\text{original salt remaining}}. \tag{2.16}$$

Either form shows $Q(t) \to 25$ as $t \to \infty$, confirming $Q_L = 25$, and that the approach is faster for larger r.

Now take $r = 3$ and $Q_0 = 2Q_L = 50$, so $Q(t) = 25 + 25e^{-0.03t}$. Since 2% of 25 is 0.5, we want $Q(T) = 25.5$:

$$25e^{-0.03T} = 0.5 \Rightarrow T = \frac{\ln 50}{0.03} \approx 130.4 \text{ min}.$$

To achieve $T = 45$ instead, set $t = 45$, $Q_0 = 50$, $Q = 25.5$ in (2.16) and solve for r:

$$r = \frac{100}{45} \ln 50 \approx 8.69 \text{ gal/min}.$$

■

Observation . Equation (2.15) is another relaxation process (Corollary 1.3.10), with $Q_{\text{eq}} = 25$ and relaxation time $\tau = 100/r$ minutes. The tank forgets its initial contents on the time scale of one "tank volume" of throughput.

Remark 2.3.3. Models of this kind are used for a pollutant in a lake or a drug in an organ of the body, not just salt in a tank. In such cases the flow rates may be hard to determine or may vary with time, the concentration may be far from uniform, and the inflow and outflow rates may differ — in which case the changing volume of liquid must also be tracked.

§2.3.ii. Seasonal Growth

In subsection 1.3.ii the coefficients were constants. Real environments vary with the seasons, which makes the coefficients *periodic functions of* t — and the techniques of ğ2.1 and ğ2.2 handle this at no extra cost.

Example 2.3.4 (A seasonally varying growth rate)**.** A population grows at a rate that varies with the season:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = (0.5 + \sin t)\,\frac{y}{5}, \qquad y(0) = 1. \tag{2.17}$$

Find the doubling time τ, and compare it with the doubling time obtained by replacing the growth rate by its average value 0.1.

Solution. Equation (2.17) is separable (it is also linear). Separating,

$$\frac{\mathrm{d}y}{y} = \frac{0.5 + \sin t}{5}\mathrm{d}t \Rightarrow \ln y = \frac{0.5t - \cos t}{5} + C,$$

and $y(0) = 1$ fixes $C = 1/5$. Hence

$$y(t) = \exp\left(0.1t + \frac{1 - \cos t}{5}\right) = \exp\left(0.1t + 0.2\,(1 - \cos t)\right). \tag{2.18}$$

The doubling time solves $0.1\tau + 0.2(1 - \cos\tau) = \ln 2$, which has no closed-form solution; numerically $\tau \approx 6.73$. Replacing the growth rate by its average 0.1 gives instead $y = e^{0.1t}$ and $\tau_{\text{avg}} = \ln 2/0.1 \approx 6.93$. ■

Observation . Two things are worth noticing.

(i) Because (2.18) is y_0 times a function of t alone, the doubling time does *not* depend on the initial population. This is special to the linear growth law; it fails as soon as crowding is modelled.

(ii) The seasonal solution is not the averaged solution: the oscillating term $0.2(1 - \cos t)$ is never negative, so (2.18) runs *ahead* of $e^{0.1t}$ and doubles sooner. Averaging the coefficients of a differential equation is not the same as averaging its solutions.

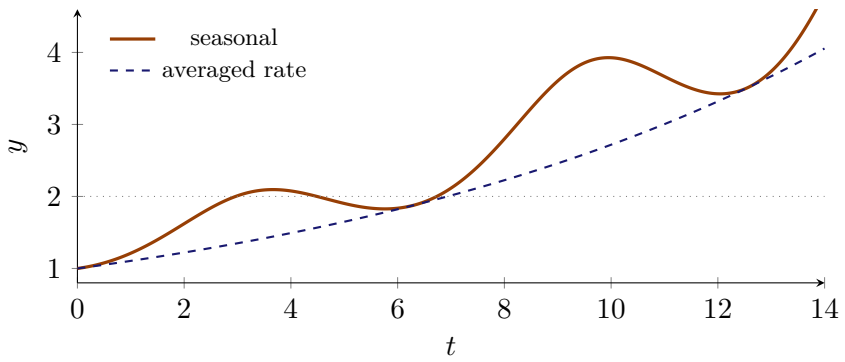


Figure 2.4.: Solution (2.18) against the constant-rate solution $e^{0.1t}$. The seasonal population crosses $y = 2$ at $\tau \approx 6.73$, slightly before $\tau_{\text{avg}} \approx 6.93$.

The same idea applies when it is the *input* that varies seasonally rather than the growth rate.

Example 2.3.5 (Chemicals in a pond)**.** A pond initially contains 10 million gallons of fresh water. Water containing an undesirable chemical flows in at 5 million gal/year and the mixture flows out at the same rate. The incoming concentration varies periodically: $\gamma(t) = 2 + \sin(2t)$ g/gal. Find the amount of chemical in the pond at any time.

Solution. The volume stays at 10^7 gal. Let t be measured in years and $Q(t)$ in grams. By the same balance law (2.14),

$$\text{rate in} = (5 \times 10^6)\,(2 + \sin(2t)) \ \text{g/yr}, \qquad \text{rate out} = (5 \times 10^6) \cdot \frac{Q(t)}{10^7} = \frac{Q(t)}{2} \ \text{g/yr},$$

so

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = (5 \times 10^6)\,(2 + \sin(2t)) - \frac{Q(t)}{2}.$$

To tame the coefficients, set $q(t) = Q(t)/10^6$ (so q is measured in megagrams, i.e. metric tons). Every term then carries a factor 10^6, which cancels:

$$\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{1}{2}q = 10 + 5\sin(2t), \qquad q(0) = 0. \tag{2.19}$$

This is linear with constant coefficient, so $\mu(t) = e^{t/2}$, and integrating gives the general solution

$$q(t) = 20 - \frac{40}{17}\cos(2t) + \frac{10}{17}\sin(2t) + ce^{-t/2}.$$

The initial condition requires $c = -300/17$, so

$$q(t) = 20 - \frac{40}{17}\cos(2t) + \frac{10}{17}\sin(2t) - \frac{300}{17}e^{-t/2}. \tag{2.20}$$

■

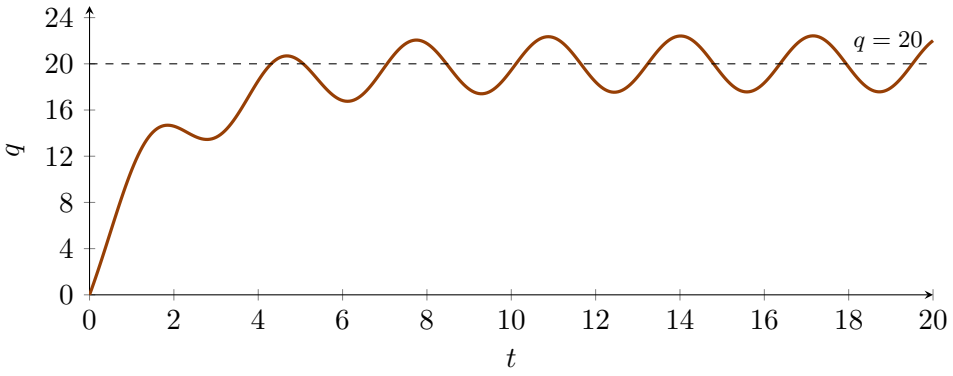


Figure 2.5.: Solution (2.20) of the pond problem $q' + q/2 = 10 + 5\sin(2t)$, $q(0) = 0$.

Observation . The exponential term matters for small t but decays rapidly; thereafter the solution oscillates about the constant level $q = 20$. Were the $\sin(2t)$ term absent from (2.19), $q = 20$ would be exactly the equilibrium solution. With it, the system settles not onto a point but onto a *periodic* response of the same frequency as the forcing — the **steady state** — while the part that remembers the initial condition, the **transient**, dies out on the relaxation time $\tau = 2$ years.

Remark 2.3.6. This model assumes the water level is controlled entirely by the flows — no evaporation, seepage or rainfall — that no chemical is absorbed by fish or other organisms, and that the concentration is uniform throughout the pond.

§2.3.iii. Compound Interest

Example 2.3.7 (Continuous compounding)**.** A sum S_0 is deposited in an account paying interest at annual rate r, compounded *continuously*. Find the value $S(t)$.

Solution. The rate of change of the investment equals the rate at which interest accrues, which is r times the current value:

$$\frac{\mathrm{d}S}{\mathrm{d}t} = rS, \qquad S(0) = S_0.$$

This equation is both linear and separable, and gives

$$S(t) = S_0 e^{rt};$$

a continuously compounded account grows exponentially. ■

If deposits or withdrawals also occur at a constant rate k (positive for deposits, negative for withdrawals), the model becomes

$$\frac{\mathrm{d}S}{\mathrm{d}t} - rS = k, \tag{2.21}$$

which is linear with integrating factor e^{-rt}, so $S(t) = ce^{rt} - k/r$. The initial condition forces $c = S_0 + k/r$, whence

$$S(t) = \underbrace{S_0 e^{rt}}_{\text{return on the initial amount}} + \underbrace{\frac{k}{r}\left(e^{rt} - 1\right)}_{\text{due to deposits/withdrawals}}. \tag{2.22}$$

Example 2.3.8 (A retirement account)**.** Someone opens a retirement account at age 25 and invests \$2000 per year continuously thereafter, at a rate of return of 8%. What is the balance at age 65?

Solution. Here $S_0 = 0$, $r = 0.08$, $k = 2000$, and we want $S(40)$. By (2.22),

$$S(40) = \frac{2000}{0.08}\left(e^{3.2} - 1\right) = 25{,}000\left(e^{3.2} - 1\right) \approx \$588{,}313.$$

The total amount invested is only \$80,000; the remaining \$508,313 is accumulated return. The result is quite sensitive to r: with $r = 0.075$ one gets \$508,948, and with $r = 0.085$, \$681,508. ■

Remark 2.3.9 (Continuous versus discrete compounding)**.** If interest is compounded m times per year, then

$$S(t) = S_0\left(1 + \frac{r}{m}\right)^{mt},$$

and the connection with the continuous model is the familiar limit

$$\lim_{m\to\infty} S_0\left(1 + \frac{r}{m}\right)^{mt} = S_0 e^{rt}.$$

In practice the frequency of compounding matters little: at $r = 8\%$, over ten years the difference between quarterly and continuous compounding is about \$17.50 per \$1000 invested, i.e. under \$2 per year. (The annual yield is 8.24% for quarterly and 8.33% for daily or continuous compounding.)

> **Notation .** Since the same model applies to investments accruing dividends and capital gains as well as interest, we refer to r as the **rate of return**.

§2.3.iv. Escape Velocity

Example 2.3.10 (Escape velocity)**.** A body of constant mass m is projected away from the earth perpendicular to its surface with initial velocity v_0. Ignoring air resistance but taking into account the variation of gravity with distance, find the velocity during the ensuing motion, the initial velocity needed to reach a maximum altitude $A_{\max}$, and the least initial velocity for which the body never returns — the **escape velocity**.

Solution. Let the positive x-axis point away from the centre of the earth along the line of motion, with $x = 0$ on the surface and R the radius of the earth. The gravitational force (the body's weight) is inversely proportional to the square of the distance from the earth's centre:

$$w(x) = -\frac{k}{(x+R)^2}.$$

At the surface $w(0) = -mg$, so $k = mgR^2$ and $w(x) = -mgR^2/(R+x)^2$. Since no other force acts, Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{mgR^2}{(R+x)^2}, \qquad v(0) = v_0. \tag{2.23}$$

Unfortunately (2.23) involves *three* variables t, x and v. The trick is to eliminate t by taking x as the independent variable; by the chain rule,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x},$$

so that

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{gR^2}{(R+x)^2}. \tag{2.24}$$

Equation (2.24) is separable (but not linear); separating and integrating,

$$\frac{v^2}{2} = \frac{gR^2}{R+x} + c.$$

Since $x = 0$ when $t = 0$, the initial condition becomes $v = v_0$ at $x = 0$, giving $c = v_0^2/2 - gR$ and

$$v = \pm\sqrt{v_0^2 - 2gR + \frac{2gR^2}{R+x}}, \tag{2.25}$$

with the plus sign while the body rises and the minus sign while it falls back.

Note that (2.25) gives velocity as a function of *altitude*, not of time. Setting $v = 0$ and $x = A_{\max}$ and solving,

$$A_{\max} = \frac{v_0^2 R}{2gR - v_0^2}, \qquad \text{equivalently} \qquad v_0 = \sqrt{2gR\,\frac{A_{\max}}{R + A_{\max}}}.$$

Letting $A_{\max} \to \infty$ gives the escape velocity

$$v_e = \sqrt{2gR} \approx 11{,}1\,\mathrm{km\,s^{-1}} \approx 6.9\ \mathrm{mi/s}. \tag{2.26}$$

■

Observation . The trick used here — trading the independent variable t for the state variable x — is worth remembering. It works whenever the force depends on position but not explicitly on time, and it is the differential-equations shadow of conservation of energy: multiplying $m\ddot{x} = F(x)$ by $\dot{x}$ and integrating gives $\frac{1}{2}mv^2 - \int F = \text{const}$, which is what

> equation (2.25) says.

Remark 2.3.11. This calculation neglects air resistance, so the true escape velocity is somewhat higher. On the other hand, launching from high altitude reduces both gravitational and frictional forces (air resistance in particular falls off rapidly with altitude). Note too that it is impractical to impart such a large velocity instantaneously; real space vehicles accelerate over a period of minutes.